

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination Nov/Dec 2015

PH204 Pharmaceutical Analysis - I

Date: 26.11.2015, Thursday

Time: 10:00a.m to 1:00p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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| A | [i] Write a note on Law of Mass Action | 07 |
| | [ii] Explain Common Ion Effect | 03 |
| B | [i] Define: (a) pH, (b) Buffer, (c) Indicator, (d) Buffer capacity | 04 |
| | [ii] Explain the Theory of Acid Base Indicators. | 06 |
| C | [i] Comment: EDTA is widely used for Complexometric Titrations | 03 |
| | [ii] What are the ideal requirements for Metal Indicators to be used for Complexometric Titrations | 03 |
| | [iii] Explain the different types of Titrations involved in Complexometric Titrations. | 04 |

Q 2 Attempt any TWO of the following

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|----------|--|----|
| A | Write a note on the steps involved in Gravimetric Analysis of a Substance. | 10 |
| B | [i] Define Co-precipitation and explain the co-precipitation mechanisms. | 08 |
| | [ii] Define Saturation and Supersturation. | 02 |
| C | [i] Write a note on Continuous Extraction Technique with suitable diagram. | 06 |
| | [ii] Enlist the factors affecting Extraction of a compound. | 04 |

SECTION- II

Q 3 Attempt any TWO of the following

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|----------|---|----|
| A | [i] Give the classification of analytical techniques. | 05 |
| | [ii] Define calibration and explain the different types of calibration. | 05 |

B	[i] Calculate the molarity of 100g KOH in 1 litre of solvent. (Mol. Wt of KOH = 56 gmol ⁻¹)	02
	[ii] Calculate the normality of 100g HCl in 1 litre of solvent. (Mol. Wt of HCl = 36.5 gmol ⁻¹)	02
	[iii] Explain factors affecting Precipitation Titration.	06
C	[i] Define Precipitation Titration. What are the requirements for Precipitation Titration.	04
	[ii] Explain Fajan's Method of Precipitation Titration.	06
Q 4	Attempt any TWO of the following	
A	[i] Define Non-Aqueous Titrations. Why are Non-Aqueous Titrations preferred.	03
	[ii] Write a note on leveling and differentiating effect observed in non-aqueous titrations giving suitable example.	07
B	[i] Define (a) Oxidation number and (b) Self Indicator	02
	[ii] Calculate the Redox Potential of the following cell	03
	$\text{Zn} \text{ZnSO}_4 \text{CuSO}_4 \text{Cu}$	
	$E^0_{\text{Zn} \text{Zn}^{2+}} = -0.76\text{V} \quad \text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$	
	$E^0_{\text{Cu}^{2+} \text{Cu}} = 0.34\text{V} \quad \text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$	
	[iii] Write a note on Iodimetry and Iodometric titrations.	05
C	[i] Write a note on types of redox indicators with examples.	05
	[ii] What is Karl Fisher Titration? Give the Chemical reactions involved and the significance of each reagent in Karl Fisher titration.	05